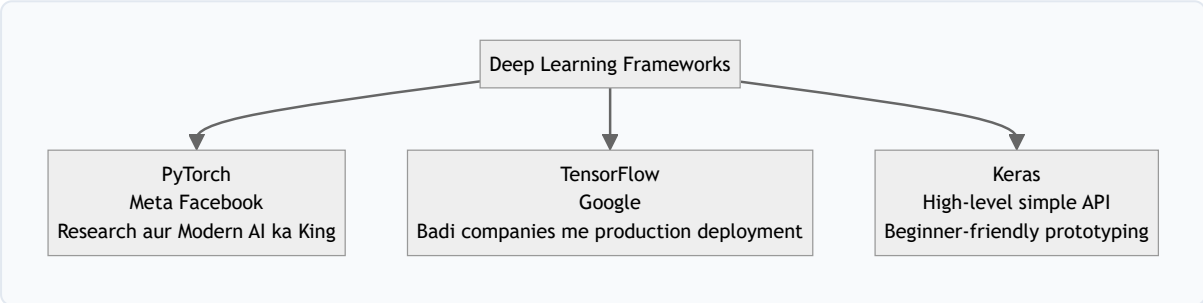
Submodule 36.06: PyTorch vs TensorFlow vs Keras (Hinglish)

1. Deep Learning Frameworks ki Zaroorat

Agar hum raw Python ya NumPy se neural network banayein, toh calculus ke derivatives aur gradients manually likhne padenge jisme hazaron lines ka code lag jayega. Deep learning frameworks **Automatic Differentiation (Autograd)** provide karte hain aur calculations ko GPU par bohot tezi se chalate hain.



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2. Comparison Matrix

Feature	PyTorch	TensorFlow	Keras
Company	Meta (Facebook)	Google	Google (ab TF ke andar integrated)
Graph Style	**Dynamic**: Code line-by-line execute hota hai	Pehle static tha, ab TF 2.0 me eager execution hai	High-level abstraction
Coding Style	Bilkul aam **Python** jaisa lagta hai	Thoda complex aur enterprise focused	Bohot simple, 4-5 lines me model ban jata hai
Kahan Use Hota Hai	AI Research, ChatGPT, LLMs, Hugging Face	Badi companies ke production servers aur mobile apps me	Shuruaat seekhne aur rapid prototyping ke liye

Feature	PyTorch	TensorFlow	Keras
Debugging	Bohot easy (normal `print` aur break points chalte hain)	Thoda mushkil	Aasan

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3. Hum PyTorch kyun padh rahe hain?

1. **Python jaisa natural**: PyTorch likhte waqt lagta hai ki hum normal Python code likh rahe hain.
2. **Debugging aasan hai**: Dynamic graph hone ki wajah se kisi bhi line par `print()` karke tensor check kar sakte hain.
3. **Industry aur Research Standard**: Aaj 80%+ nayi AI research aur open-source models (jaise Llama, Stable Diffusion) PyTorch me hi bante hain.

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4. Visual Reference

